

Experiment 1: Linux Command Execution Using `fork()` and `exec()`

Course: Operating Systems and Systems Programming (OSSP)

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Date: 31/7/2026

OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Experiment Steps:

Step 1: Check Whether GCC is Installed

To check if the GNU C Compiler (GCC) was installed, I executed the following command:

```
gcc --version
```

```
charishma@LAPTOP-JVQGRAD3:~$ gcc
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
```

Observation:

The GCC compiler was successfully installed on the system. The command displayed the installed GCC version as **13.3.0**, confirming that the compiler is ready to compile C programs.

Step 2: Navigate to the Program Directory

The directory containing the C program was opened using:

```
charishma@LAPTOP-JVQGRAD3:~$ cd "/mnt/c/Users/Charishma/Downloads/EXP"
```

To verify the file, the following command was executed:

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ ls
```

Observation:

The terminal displayed:

```
exp1.c
```

This confirmed that the source file was present in the directory.

Step 3: Compile the Program

The C program was compiled using the GCC compiler with the following command:

```
charishma@LAPTOP-JVQGRAD3:~$ gcc exp1.c -o exp1
```

Observation:

The program compiled successfully without any errors, and an executable file named **exp1** was created.

Step 4: Execute the Program

The compiled program was executed using:

```
./exp1
```

When prompted:

Enter Linux Command

a Linux command was entered, and the program executed it successfully using the `system()` function.

Step 5: Execute Different Linux Commands Step

1) Ls

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ ./exp1
Enter Linux command: ls
Parent Process
Parent PID = 985
Child PID = 986
Child Process
Child PID = 986
exp1 exp1.c sample.txt
Child process completed.
```

2) cd

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ ./exp1
Enter Linux command: cd
Parent Process
Child Process
Parent PID = 1009
Child PID = 1016
Child PID = 1016
Command execution failed.
Child process completed.
```

3) Uname

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ ./exp1
Enter Linux command: uname
Parent Process
Parent PID = 1026
Child PID = 1031
Child Process
Child PID = 1031
Linux
Child process completed.
```

4) Lscpu

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          46 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 18
On-line CPU(s) list:    0-17
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) Ultra 5 125H
CPU family:             6
Model:                 170
Thread(s) per core:     2
Core(s) per socket:     9
Socket(s):              1
Stepping:               4
BogoMIPS:               5990.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge
                        e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_t
                        nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx
                        c movbe popcnt tsc_deadline_timer aes xsave avx f16c
                        fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
                        1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
                        1 xsave avx_vnni vnni umip waitpkg gfni vaes vpclmu
                        ear serialize ibt flush_lld arch_capabilities
```

5) Lsblk

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
```

7)Top

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ top
top - 09:39:42 up 17 min, 1 user, load average: 0.03, 0.03, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7685.2 total, 6966.8 free, 633.3 used, 237.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7051.8 avail Mem
```

8) whoami

```
charishma@LAPTOP-JVQGRAD3:/mnt/c/Users/Charishma/Downloads/EXP$ ./exp1
Enter Linux command: whoami
Parent Process
Parent PID = 996
Child PID = 997
Child Process
Child PID = 997
charishma
Child process completed.
```

